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PROJECT PROPOSAL: TEACHER MANAGEMENT SYSTEM

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1.0 Project background

Tadika Genius Aulad is an established, multi-class early childhood education center with a growing student enrollment and an expanding team of teaching and administrative staff. Currently, the daily operations, human resource records, and facility issues of the kindergarten are managed with bits of social messaging apps such as WhatsApp with 9 teachers and 1 principal in a group. While these methods sufficed when the center was smaller, the increasing scale of the kindergarten has caused high administrative overhead and operational inefficiencies that divert resources from student development.

1.1 Problem Statement

Efficiency and accuracy in communication are essential in early childhood education to create a safe and supportive environment for children. Tadika Genius Aulad, along with many modern kindergartens, have to deal with a variety of administrative functions, including teacher certification, teacher performance evaluation, duty schedule and maintenance report. Currently, much of this work is being done through manual methods or disjointed means of communication, which can result in information gaps, delays in responding to kindergarten maintenance and repair issues, and a poor ability to monitor teachers' long-term professional growth:

- **Inadequate Workload Allocation**

Unable to equitably allocate daily tasks and monitor their progress in real-time.

- **Poor Communication**

Difficulty in timely reporting maintenance problems (e.g., property damage) or bullying issues, which need to be addressed by the administration.

- **Isolated Data**

Inability for Principals to centrally see an overview of teacher performance (KPIs), training and leave balances.

1.2 Objectives

The primary goal of this project is to design and develop a centralized Teacher Management System (TMS) to transition Tadika Genius Aulad into a seamless, digital-first ecosystem. To achieve this, the project will fulfill the following specific objectives:

- **To Optimise Processes**

Automate on-call rosters and holiday requests to promote equitable workloads.

- **To Improve Accountability**

Offer a transparent system for evaluating performance (KPI) and training.

- **To Streamline Response**

Offer a real-time reporting system for incidents and maintenance with photos and push alerts.

- **To Consolidate Information**

Establish a digital and secure database for teacher management and certificates.

2.0 Target users

This system is designed for two main target user groups of a total of 10 users, consisting of 1 principal, 1 kindergarten owner (administrator), and 8 teachers.

1. Administrator (Principal & Kindergarten Owner)

- Manage teacher records
- Create new teacher accounts
- Assign tasks to teachers
- Approve leave applications
- Evaluate teacher performance (KPI)
- Post and manage training programmes
- Track training participation
- Assign training to teachers if no one joins
- Review maintenance reports submitted by teachers
- Take appropriate action and update report status until resolved

2. Teachers

- View assigned tasks
- View performance summaries
- Apply for leave
- Submit maintenance reports
- Register or join training programmes

- Post training reflection after completing training
- Update personal information

3.0 Mandatory modules

This section outlines the mandatory modules and operational responsibilities that extend beyond the six industry-standard modules requested.

3.1 User Authentication and Profile

This module includes three baseline modules, which are login, registration and managing users.

Login Module

The Login Module provides a secure gateway for authenticated users to access the platform. It ensures data protection by verifying user credentials against the system database.

- Secure Authentication: Users log in using their registered email address and password.
- Role-Based Redirection: Upon successful authentication, the system automatically detects the user's role (Admin or Teacher) and directs them to their respective dashboard interface.
- Security Safeguards: Includes password visibility toggles, "Forgot Password" self-service recovery links, and account lockout protocols after consecutive failed login attempts to prevent unauthorized access.

Registration Module

The Registration Module allows new users to create an account on the platform, establishing their initial profile and system identity.

- Account Creation Form: Collects essential user information, including personal details, qualifications, supporting documents, and securely hashed password information.
- Identity Verification: Employs email verification via OTP to ensure account authenticity before granting system entry.
- Approval Queue (Optional/Configurable): New registrations are held in a pending state until verified and by admin, preventing unauthorized external access.

User Management Module

The User Management Module provides administrators with a centralized control panel to supervise all system accounts, enforce security policies, and maintain data integrity.

- **User Directory & Profiles:** A searchable, filterable master list of all registered personnel, displaying their current status (Active, Inactive, Pending), and personal details.
- **Account Lifecycle Controls:** Provides administrative actions to deactivate or archive user accounts. Admins can also trigger manual password resets or edit profile details on behalf of staff members when necessary.

3.2 User Interface (UI) and Navigation

This section focuses on the visual design and structural layout of the Teacher Management System. It ensures that the application is professional, user friendly and provides a seamless user experience across different users.

3.3 Data Configuration and Management

This section is to set up connection to the database used by the system, Firebase. It consolidates the data entities needed for the entire system and defines the relationships between them to ensure data consistency and integrity across the application.

3.4 Push Notifications and Communication

Push Notifications send real time alerts to both teachers and admins about tasks, training, reports and updates from the organisation. This helps everyone stay informed and respond quickly without constantly checking the system.

4.0 Industry Modules

This section contains six modules and their respective responsibilities required in the project.

4.1 Teacher Records

The Teacher Records Module is designed to manage and maintain all teacher-related information. This module enables teachers to securely update and manage their personal information, qualifications and supporting documents through an online platform. At the same time, administrators are able to verify, approve, monitor and manage teacher records efficiently.

In addition, uploaded documents are validated against the information entered by the teacher to ensure data accuracy.

1. Teacher Functions

Teachers are able to:

- Login using email and password
- Update personal information
- Reupload supporting documents if the uploaded files are invalid, unclear, incomplete or blurred.
- View document verification status

Teachers are only allowed to edit:

- Contact number
- Residential address
- Email address
- Emergency contact person and number
- Marital status

However, important identity information cannot be modified by teachers after registration to maintain data integrity and prevent unauthorized changes. These non-editable information include:

- Full name
- IC Number
- Gender
- Date of Birth

If changes to identity information are required, teachers must submit a request to the administrator together with supporting documents for manual verification and approval.

2. Administrator Functions

Administrators are responsible for managing and verifying teacher records. The functions available to administrators include:

- View all teacher records
- Approve or reject teacher records
- Edit teacher information if necessary
- Verify uploaded documents
- Track teacher verification status

- Monitor document submission completeness

3. Teacher Information Collected

Field	Validation Rule
Full Name	Cannot contain numbers
IC Number	Must follow Malaysian IC format
Gender	Must select either Male or Female
Date of Birth	Cannot be a future date
Address	Cannot be empty
Phone Number	Must contain digits only and valid phone number format
Email Address	Must be a valid email format
Marital Status	Must select from Single, Married, Divorced or Widowed
Emergency contact person	Cannot contain numbers
Emergency contact number	Must contain digits only and valid phone number format

4. Supporting Documents Required

All documents are only allowed in PDF, JPG and PNG formats and maximum 5MB per file.

- Copy of Identification Card (MyKad)
- Passport photo
- Resume or Curriculum Vitae (CV)
- Latest Academic Certificates
- Medical Check Up Report
- Header of Bank Statement

5. Data Validation and Verification

The system uses Optical Character Recognition (OCR) technology such as on-device ML kit OCR to extract text and information from uploaded documents such as MyKad, academic certificates and bank statements headers. The extracted information is then compared with the information entered by the teacher in the system.

The validation and verification process includes:

- Matching teacher full name with MyKad, certificates and bank statement
- Verifying IC Number against uploaded MyKad
- Checking date of birth consistency with identification documents
- Verifying qualification details using academic certificates
- Validating bank account holder information through bank statement header
- Ensuring uploaded medical reports are valid and readable

If inconsistencies or mismatched information are detected, the system will display a warning message.

4.2 Leave Management

The Leave Management Module enables teachers to apply for leave while allowing administrators to review and track applications against specific quotas. The system ensures transparency by automatically managing balances for these categories, including monthly allocations for half-day leaves. The system automatically handles application rules, mandatory documentation requirements, and user-specific actions based on the 10 distinct leave types as configured in the Table 1 below:

Table 1: Specification for each type of leave

Leave Type	Quota (Days)	Mandatory Supporting Documents
Annual Leave	8	None (Optional remarks)
Medical Leave (MC)	14	Medical Certificate (MC) from a clinic/hospital
Unpaid Leave	8	Justification letter/document
Maternity Leave	98	Hospital admission letter / Certified medical report
Marriage Leave	5	Marriage certificate / Wedding invitation card
Compassionate Leave	2	Death certificate / Official notice
Umrah Leave	14	Travel itinerary / Flight booking confirmation
Haji Leave	40	Official pilgrim allocation letter (e.g., Tabung Haji)
Birthday Leave	1	None (System validates via IC/Birth Date)

Half Day Leave	2 (per month)	None (Optional remarks)
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For teachers the system will prevent submission if a required mandatory document is missing and once the document is submitted, teachers can track the real-time review status of their application. As for administrator, a centralized dashboard will aggregate all pending leaves where the system will automatically recalculate the teacher's remaining balance and push an immediate real-time status notification to the teacher's device.

4.3 Task and Duty Management

This module allows teachers and administrators to track their daily tasks and duties. The system should be able to automatically generate duty schedules and assign duties fairly to teachers. Some examples of daily duties include cleaning duty, arrival duty, dismissal duty and half-full day transition duty, while weekly duties include assembly duty. Each duty has a list of tasks and the task progress is tracked. For example, cleaning duty in a classroom includes tasks such as sweeping and mopping the classroom floor, wiping and arranging the students' tables and chairs, organising teaching materials, toys, books and files and so on. For assembly duty, each week has a theme with its respective sub themes, each teacher shall be responsible for tasks such as introduction, song, islamic content, and words of the week and sight words.

Teachers can update task completion and swap duties with their colleagues. Administrators can edit the duties list, duty information and task list. The application should be able to send reminders or notifications for each task. Each features are explained in detailed below:

1. Automatically generate duty schedules

The system would assign jobs to the teachers with "Active" status and are not on leave for the day. The scheduled task would be one week ahead and updates whenever there are changes in teachers status and leave. The duty schedule would ensure that there is no overlapping of tasks at the same time for the same teacher. Besides, the system would ensure that there will be an assigned teacher for all duties.

2. Duties and tasks progress tracking

The teachers have to update the progress of the duties assigned. For each task of the duty, the teachers have to take a photo as a proof of completion of the particular task. Once all tasks under a certain duty are completed, the duty would be marked as completed. The progress of all tasks and duties can only be updated from half an hour before till half an hour after the time frame of assigned duty.

3. Swap duties

The teachers would be able to swap the assigned duties with their colleagues. When the teacher would like to swap duties, they can only select teachers with the duties of the similar hours or the teachers that do not have other duties during those hours. The teacher that is selected to swap duty would receive a notification and have to approve the swap. On the other hand, the principal can swap the tasks of any teacher. Both the teachers would receive a notification without requiring approval regarding the duty swap. Swap duties can only be performed latest by half an hour before the duty commences.

4. Modify duties

The principle can create duties, edit duties and delete duties. For each duties, there is information including duty name, locations, time, daily or weekly duties, minimum teacher in charge per location and task list..

5. Notifications

Teachers will be notified with the assigned task 30 minutes prior to duty hours.

All teachers would be able to view all the tasks and the assigned teachers. However, the teachers can only update the task assigned to the respective teachers.

4.4 Teacher Training Tracker

This module lets both the admin and teachers share and view training information in a dynamic, interactive "Social Feed" layout. This platform allows admins to post training notices and track teacher achievements. At the same time, teachers can share their own training milestones and insights, as well as view and engage with other colleagues' updates in a familiar, social media-style feed.

Admin Feature

- Create Training Notices: Post details of upcoming training sessions, including the title, organizer, scheduled date, duration, mode (online or physical), and venue or platform.
- Assign Participants: Direct specific teachers to attend required training sessions or open the session for voluntary registration.
- Monitor Completion: View a comprehensive list of submitted reflections, certificates, and photos from teachers to verify attendance and track professional development hours.

Teachers Features

- View Training Opportunities: Receive notifications and access upcoming training notices posted by the admin to see mandatory assignments or voluntary sessions.
- Log Completed Training: Submit a detailed report for any completed training sessions. The app provides a dropdown menu for teachers to categorize their training into one of the following areas:
 - Teaching Skills
 - Child Development
 - Safety and First Aid
 - Islamic Education
 - Classroom Management
 - ICT/Technology
 - Others
- Submit Proof and Documentation: Upload necessary verification materials as complementary proof, including:
 - Certificates
 - Training Photos
- Write Reflections: Input a summary of key takeaways and personal learnings from the training session, fostering continuous professional growth and sharing knowledge with the wider team.

4.5 Teacher Performance Tracker

This module enables the administrator to evaluate and monitor teachers' performance using a structured KPI-based scoring system. The admin can record positive and negative performance observations in real time based on predefined evaluation categories as shown below:

- I. Attendance and punctuality
- II. Classroom Management
- III. Teaching Performance
- IV. Student Development
- V. Documentation and Record Keeping
- VI. Communication and Professionalism
- VII. Task & duty Responsibility
- VIII. Creativity and Initiative
- IX. Training and Self Development
- X. Discipline and SOP Compliance

Each evaluation record will automatically update the teachers' KPI score according to a weighted scoring mechanism. Positive performance records may contribute between +1 to +3 marks, while negative behaviors may deduct between -1 to -5 marks depending on seriousness. This module also supports monthly performance summaries and a centralized yearly KPI evaluation to provide a comprehensive overview of teachers' performance trends throughout the year and assist administrators in the final KPI assessment process.

Teachers are only allowed to view their summarized KPI results and performance trends, where detailed evaluation records are restricted to the administrator.

The detailed system key features are as follows:

1. Performance Recording

The administrator can record teacher performance anytime by selecting relevant criteria and performance severity. The system will automatically calculate and update the KPI score accordingly based on the weighted scoring mechanism. However, the administrator

is only allowed to deduct a maximum of 10 KPI marks per teacher per day to ensure fairness and prevent excessive penalization.

2. Automated KPI Scoring System

The system follows an automatic add and deduct scoring mechanism based on predefined severity levels. Positive behaviors will increase KPI marks while misconduct or poor performance results in mark deductions.

3. Monthly Performance Report and Trend Analysis

The system generates monthly performance summaries automatically for each teacher, including total marks, increments, deductions, category breakdown and performance trends. Teachers are able to view their monthly KPI summaries and performance trends while administrators can monitor teachers' performance progression over time.

4. Yearly KPI Evaluation

At the end of the year, the system calculates the final KPI for each teacher based on accumulated monthly performance data and trends. The final result is presented as an overall KPI score with a performance rating to assist the administrator to make better decisions regarding teacher performance, improvement plans or further actions. The administrator may review and adjust the final KPI result if necessary.

5. Centralized KPI Dashboard

The administrator can monitor all teachers' KPI results in a centralized dashboard that provides an overview of school wide performance analytics. Besides, the administrator can access teacher individual performance details through a drill-down feature for further analysis of monthly trends and detailed evaluation records.

4.6 Teacher Reporting System

The Teacher Reporting Management Module offers a secure, structured and confidential place for teachers to report incidents, report operational concerns and request administration interventions. The system enables teachers to file formal reports supplemented with photographic evidence and monitor the resolution progress in real-time via an interactive status tracker. The

module has a firm reporting mechanism that automatically sends out real-time alerts between teachers and higher management to facilitate quick response from the organization.

The system supports various kinds of reports that are tailored for the different contexts in kindergarten, as described below:

No	Report Category	Description & Scope
1	Sexual Harassment Report	To safely report any unwanted or inappropriate sexual behaviour at work.
2	Bullying Report	To report physical, emotional, verbal or online bullying between kids, teachers or parents.
3	Conflict Between Staff Report	To report arguments or problems between coworkers that need the principal's help to fix.
4	SOP Violation Report	To report when someone breaks the kindergarten's official rules, safety steps or guidelines.
5	Workload Stress Report	A safe way for teachers to tell management if they are feeling too stressed, burnt out or overwhelmed with work.
6	Teacher Misconduct Report	To report wrong, bad or unprofessional behaviour by any teacher.
7	Facility Maintenance Report	To report broken things in the school like damaged walls, broken chairs, leaking pipes or things that need fixing.
8	Teaching Material Shortage Report	To ask for more classroom items like books, stationery or teaching tools when they run out.
9	Safety Hazard Report	To report dangerous things like loose wires, wet floors or broken playground toys that could hurt the children or teacher.
10	IT/System Problem Report	To report phone app bugs, computer problems, internet issues or system errors.

5.0 Module distribution table

Module	Type	Lead Member
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User Authentication & Profile	Mandatory ▾	Tan Yi Ya
UI & Navigation	Mandatory ▾	Nur Firzana
Data Configuration and Management	Mandatory ▾	Lam Yoke Yu
Push Notification & Communication	Mandatory ▾	Teh Ru Qian
Module 1: Teacher Records Management	Industry ▾	Teh Ru Qian
Module 2: Leave Management	Industry ▾	Nuraisyah
Module 3: Task & Duty Management	Industry ▾	Lam Yoke Yu
Module 4: Teacher Training Management	Industry ▾	Tan Yi Ya
Module 5: Teacher Performance Management	Industry ▾	Goe Jie Ying
Module 6: Teacher Reporting Management	Industry ▾	Nur Firzana

6.0 Justification

These modules are split into mandatory and industry components to cover both core system functions and application-specific features. Mandatory modules support overall system operations, while industry modules represent the requirement from the industry to cover some key functional areas of the kindergarten. The module allocation is based on each team member's interests and strengths, allowing us to develop each module more effectively and creatively.

7.0 Budget Estimation

The proposed cost estimation is calculated based on the resources and technical requirements involved in developing the kindergarten management system. The budget covers manpower, system development tools, cloud and database services, interface design, testing, deployment, maintenance, security measures and contingency planning. These estimated costs are intended to support the complete development lifecycle and ensure system sustainability and reliability.

Category	Cost (RM)
Manpower Cost (RM1000 per member)	6,000
Software & Development Tools	1,500

Database & Cloud Services	2,500
UI/UX Design & Prototyping	2,000
Testing & Quality Assurance	1,500
System Maintenance & Technical Support	4,000
Deployment & Configuration	1,000
Documentation & Training Materials	800
Internet & Communication Expenses	500
Security & Backup Services	1,200
Device & Hardware Testing	1,500
Contingency Cost	2,000
Total:	24,500

8.0 Conclusion

The development of the Teacher Management System for Tadika Genius Aulad marks a significant transition from traditional, manual administrative processes to a streamlined, digital-first approach. By integrating both Mandatory Module and Industry Module, the system is able to effectively address the unique operational “pain points” of kindergarten management. The strategic division of modules ensures specialized focus and high-quality development for every feature. Once implemented, the application will resolve workload imbalances and communication gaps while empowering administrators with data-driven insights. By providing teachers with a user-friendly management tool, this project fosters a transparent and efficient environment, allowing the staff of Tadika Genius Aulad to focus their energy on student growth and excellence.